ECE0202
Embedded Processors & Interfacing + Lab

Sichuan University-Pittsburgh Institute

Topics

- Course Logistics
  - Who is this Instructor?
  - What is this course about?
  - Why should I take this class?
  - How is this course organized?
- Introduction to Embedded Systems

Instructor

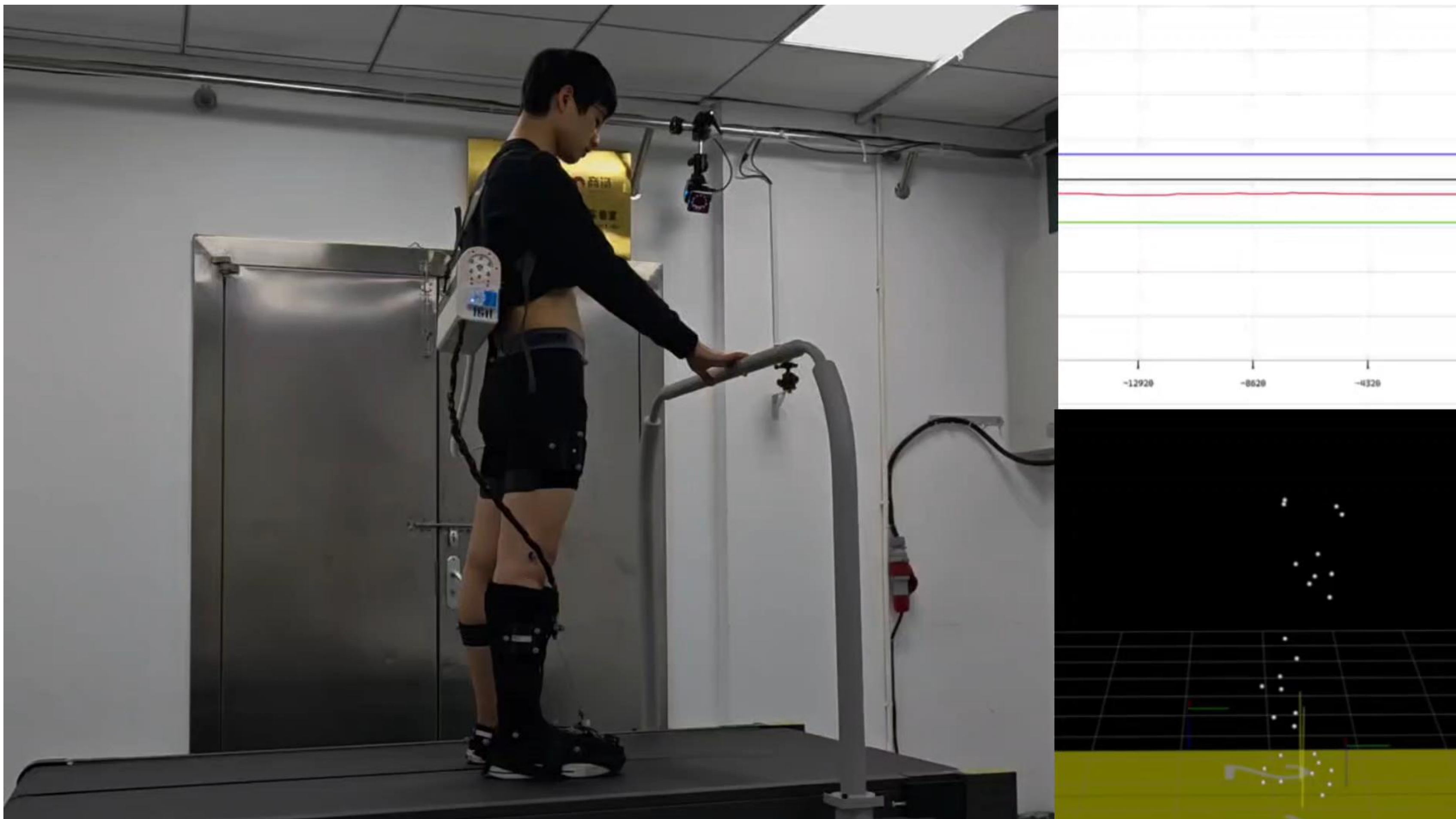
- Instructor: Dr. Fashu Xu
- Office:
  - N520, South District, Jiangan Campus
  - Rm 103, 2nd Teaching Building, East District, Huaxi Campus
- Email: xufs@scu.edu.cn
- Office Hour: By Appointment

Teaching Assistants

- Lecture1: Haiyang Yu 2023141520296@stu.scu.edu.cn
- Lab1:
  - Jiahao Li 2023141520010@stu.scu.edu.cn
  - Ruijie Ni 1595058100@qq.com
-
- Lecture2: Xiang Si 2023141520293@stu.scu.edu.cn
- Lab2:
  - Zhehang Ye 2023141520269@stu.scu.edu.cn
  - Dongyu Chen 2023141520295@stu.scu.edu.cn

What is this course about?

Ankle Exosuit in Our Laboratory Based on Embedded Systems



ECE 0202 Embedded Processors and Interfacing

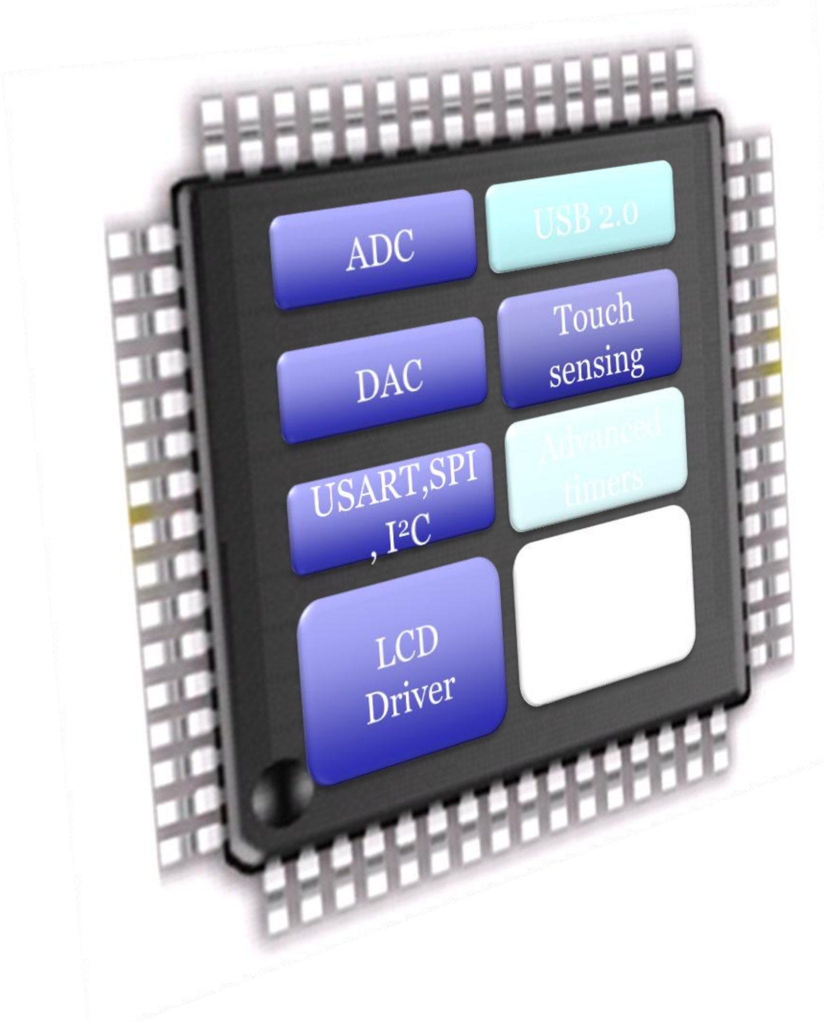
- Microprocessor based systems
- This course is the studying of how to program microprocessor to interact with physical world

Embedded Systems


KIVA PICKING
Kiva Robot

Why ARM processor

- As of 2005, **98%** of the more than one billion mobile phones sold each year used ARM processors
- As of 2009, ARM processors accounted for approximately **90%** of all embedded 32-bit RISC processors
- In 2010 alone, **6.1 billion** ARM-based processor, representing **95%** of smartphones, **35%** of digital televisions and set-top boxes and **10%** of mobile computers
- As of 2014, over 50 billion ARM processors have been produced



iPhone 5 Teardown

The A6 processor is the first Apple System-on-Chip (SoC) to use a custom design, based off the **ARMv7** instruction set.

http://www.ifixit.com

iPhone 6 Teardown

The A8 processor is the first 64-bit ARM based SoC. It supports **ARM A64, A32, and T32** instruction set.

http://www.ifixit.com

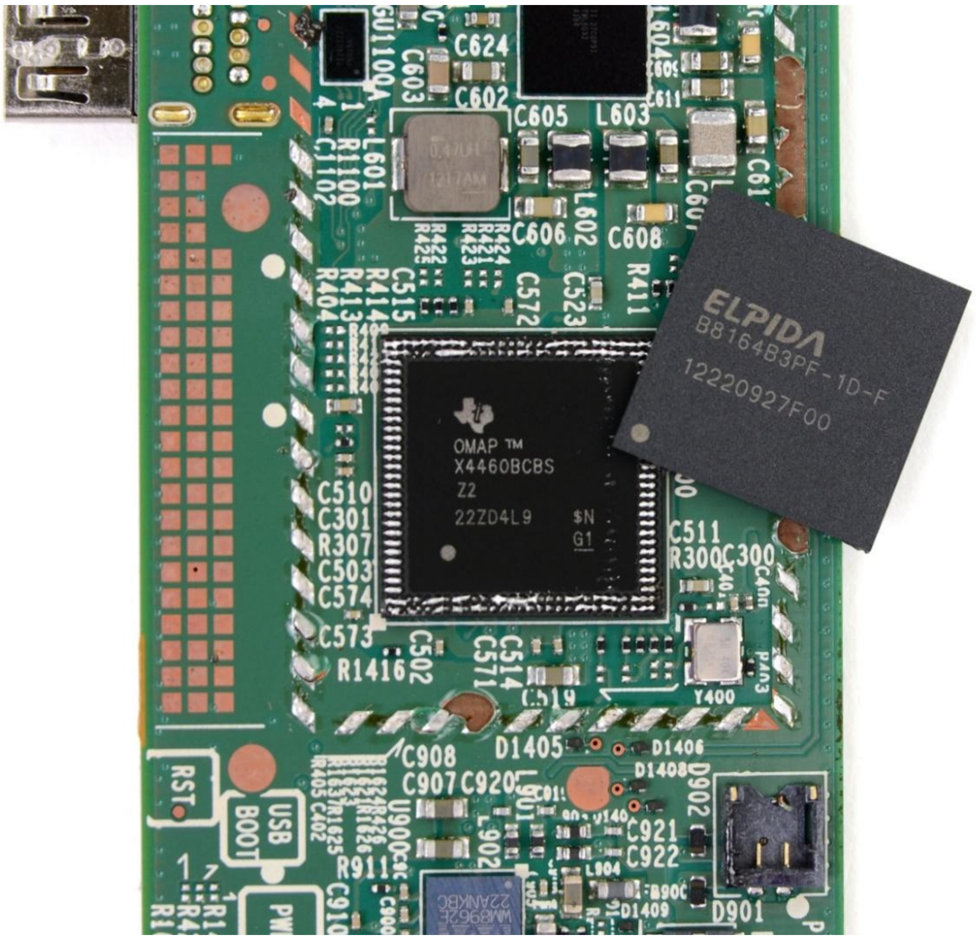
iPhone 7 Teardown

A10 processor:
- 64-bit system on chip (SoC)
- **ARMv8**-A core

Apple Watch

- Apple S1 Processor
  - **32-bit ARMv7-A** compatible
  - # of Cores: **1**
  - CMOS Technology: 28 nm
  - L1 cache 32 KB data
  - L2 cache 256 KB
  - GPU PowerVR SGX543

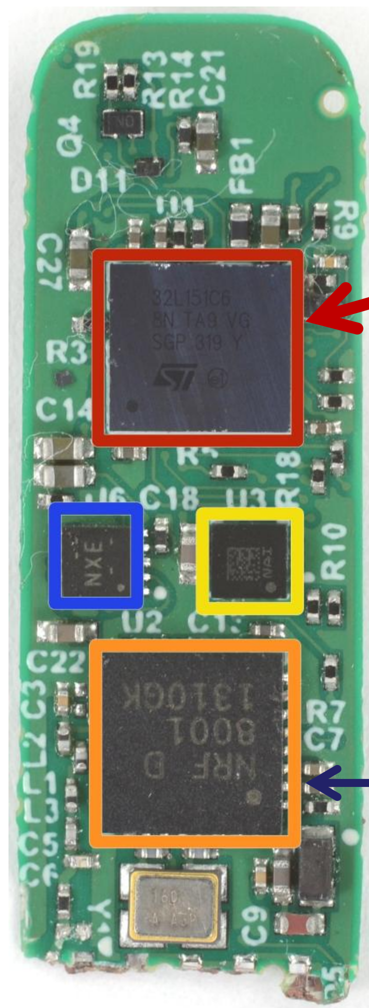
Kindle HD Fire



Texas Instruments **OMAP 4460 (Cortex-A9 + PowerVR GPU + TI Tesla DSP)** dual-core processor

http://www.ifixit.com

Fitbit Flex Teardown

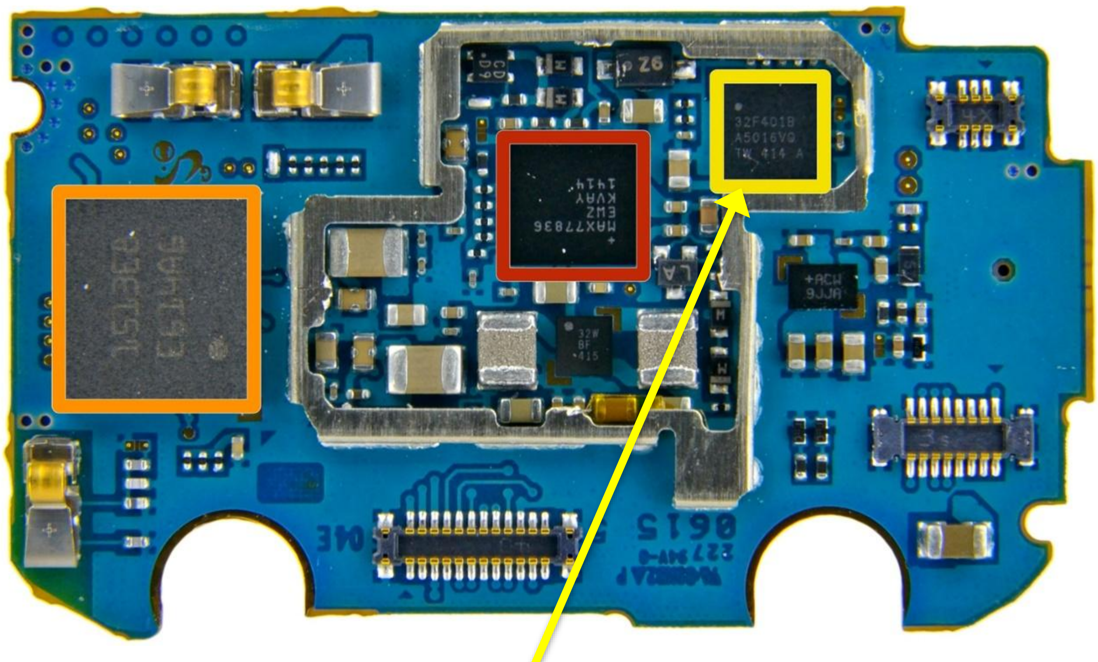


STMicroelectronics 32L151C6 Ultra Low Power ARM Cortex M3 Microcontroller

Nordic Semiconductor nRF8001 Bluetooth Low Energy Connectivity IC

http://www.ifixit.com

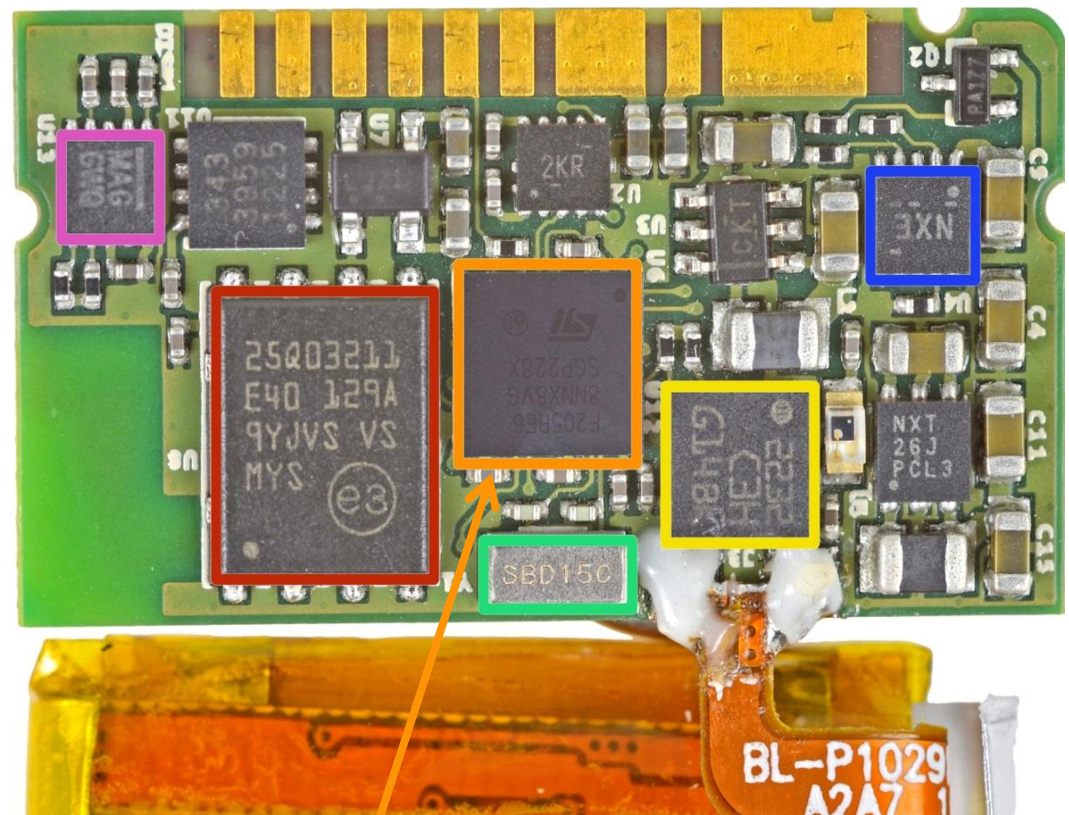
Samsung Galaxy Gear



- STMicroelectronics STM32F401B **ARM-Cortex M4** MCU with 128KB Flash

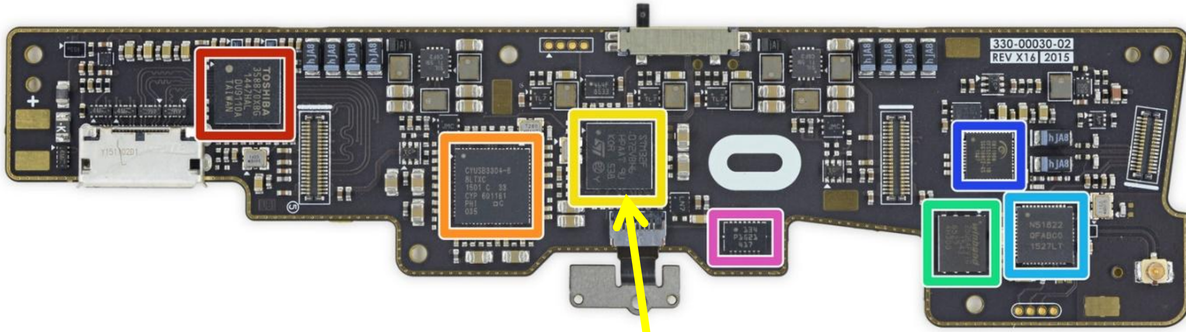
http://www.ifixit.com

Pebble Smartwatch



- STMicroelectronics STM32F205RE **ARM Cortex-M3** MCU, with a maximum speed of 120 MHz

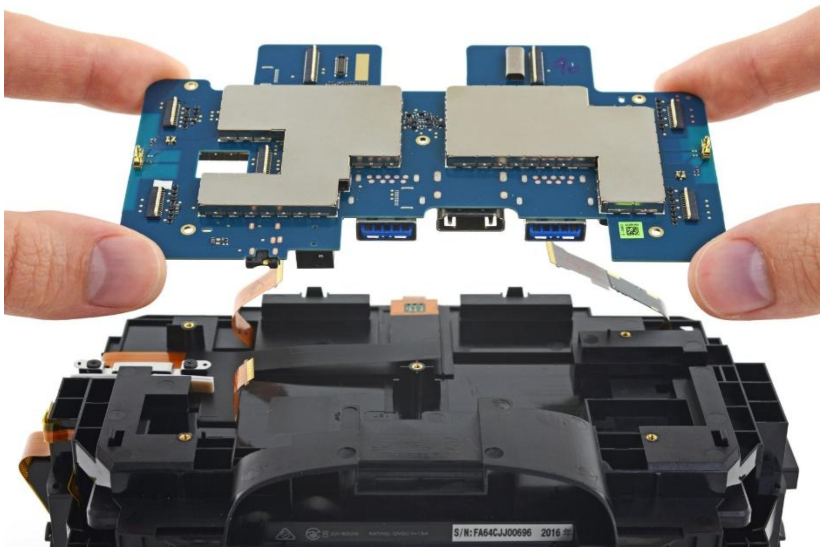
Oculus VR

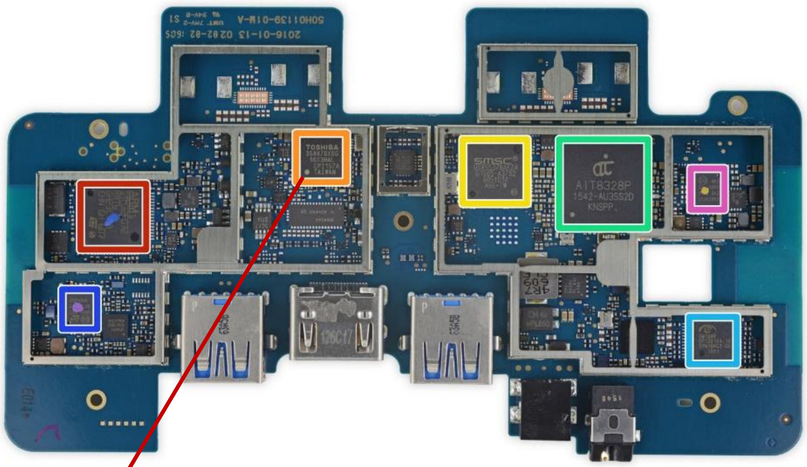


- Facebook's $2 Billion Acquisition Of Oculus in 2014
- ST Microelectronics STM32F072VB **ARM Cortex-M0** 32-bit RISC Core Microcontroller

http://www.ifixit.com

HTC Vive





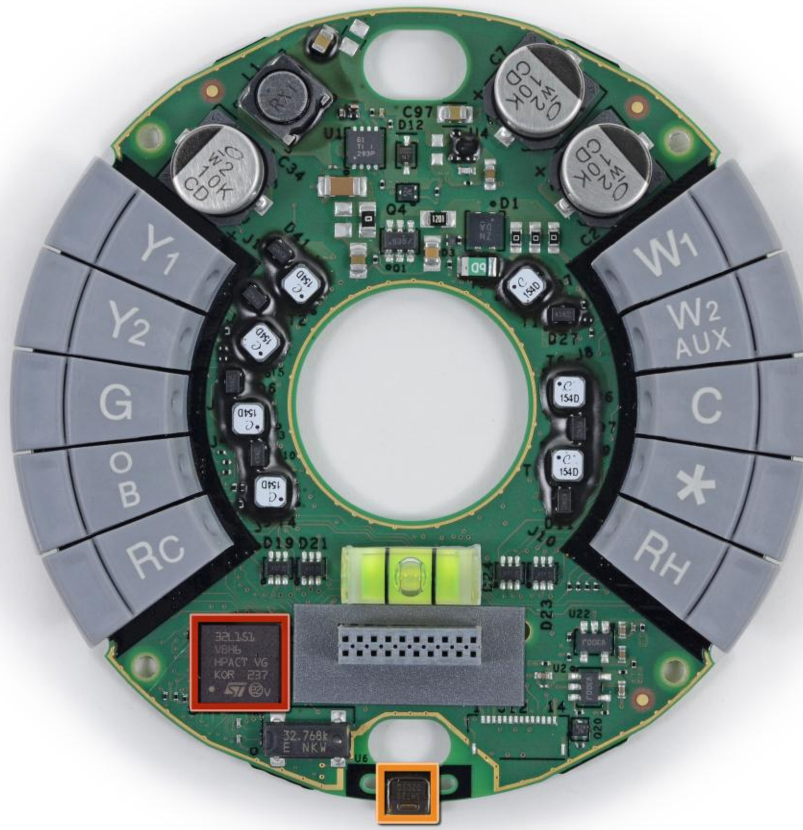
STMicroelectronics 32F072R8

ARM Cortex-M0

Microcontroller

http://www.ifixit.com

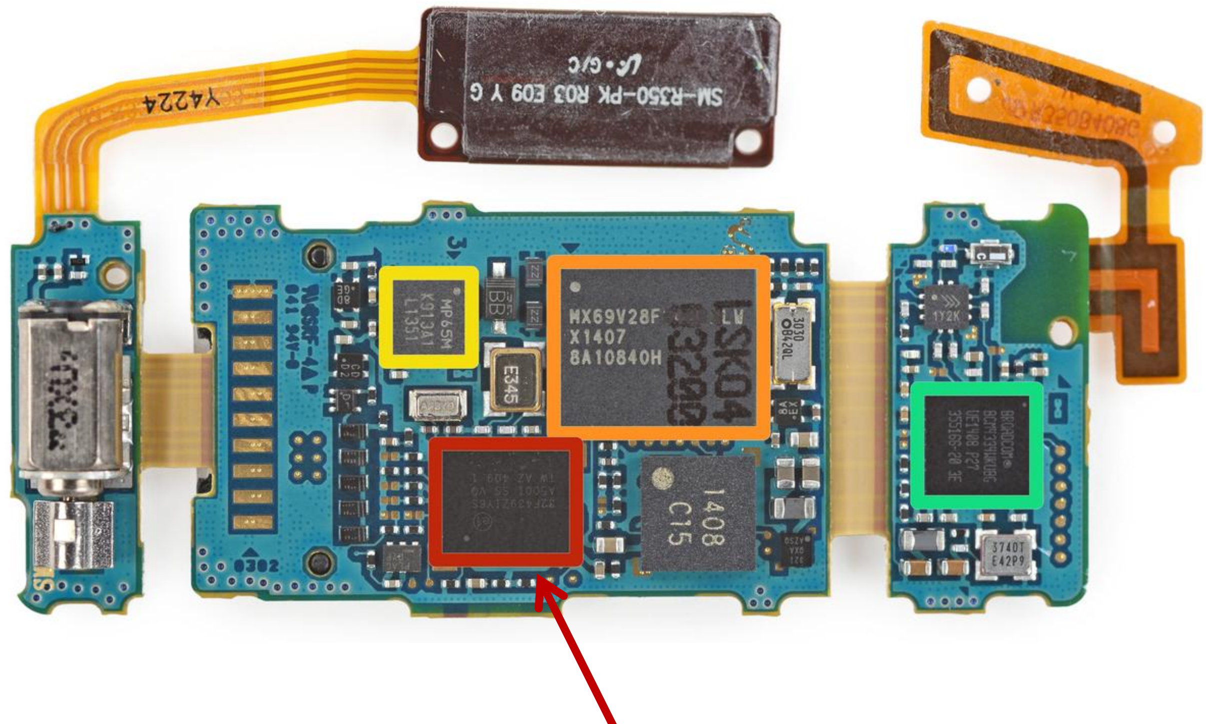
Nest Learning Thermostat



- ST Microelectronics **STM32L**151VB ultra-low-power 32 MHz ARM **Cortex-M3** MCU

http://www.ifixit.com

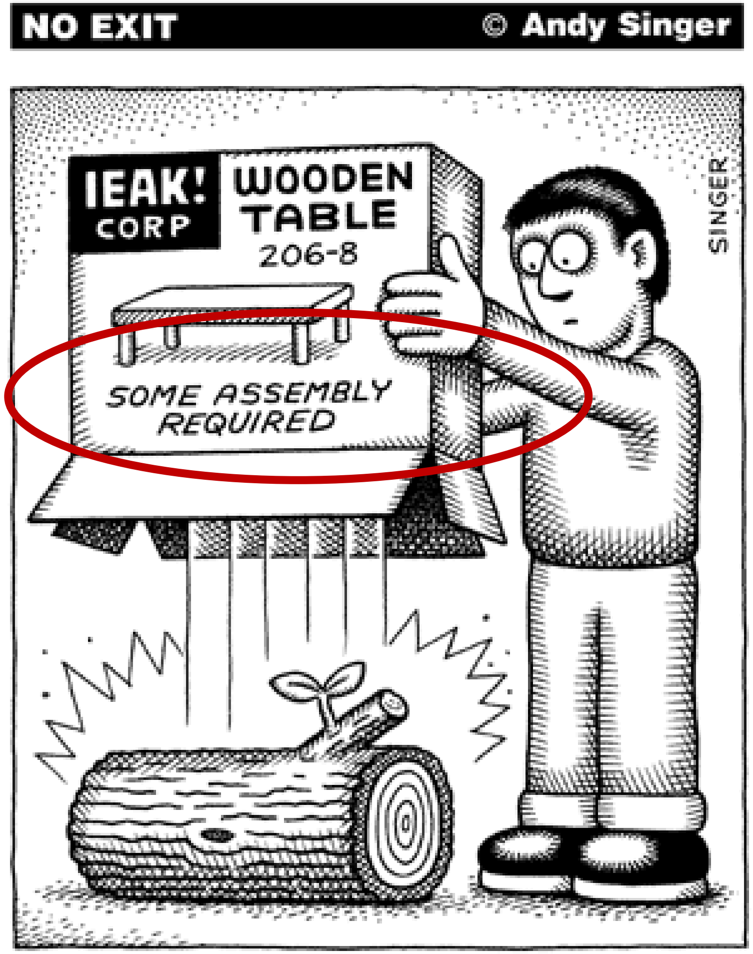
Samsung Gear Fit Fitness Tracker



- STMicroelectronics **STM32F4**39ZI 180 MHz, 32 bit **ARM Cortex-M4** CPU

http://www.ifixit.com

Assembly Programs



http://www.andysinger.com/

Why do we learn Assembly?

- Assembly isn't "just another language".
  - Help you understand how does the processor work
- Assembly program runs faster than high-level language. Performance critical codes must be written in assembly.
  - Use the profiling tools to find the performance bottle and rewrite that code section in assembly
  - Latency-sensitive applications, such as aircraft controller
  - Standard C compilers do not use some operations available on ARM processors, such ROR (Rotate Right) and RRX (Rotate Right Extended).
- Hardware/processor specific code,
  - Processor booting code
  - Device drivers
- Cost-sensitive applications
  - Embedded devices, where the size of code is limited, wash machine controller, automobile controllers
- The best applications are written by those who've mastered assembly language or fully understand the low-level implementation of the high-level language statements they're choosing.

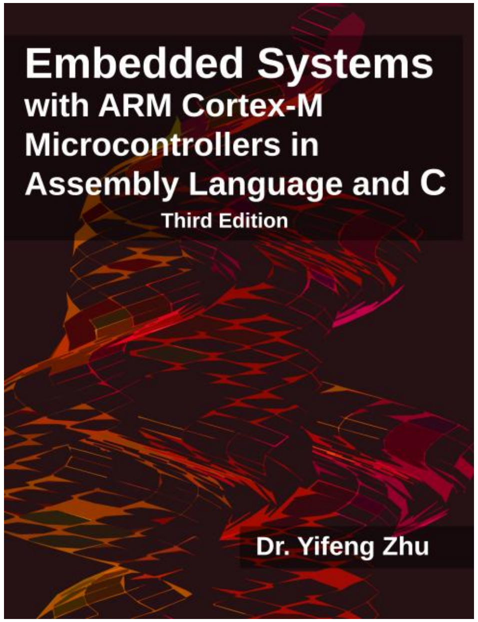
Classes

- Classroom: N209, New Building, SCUPI
- Slides
  - I will upload to Blackboard before the class

Class Structure

- Video + Lecturing
- Quizzes
- TAs will be in charge of the lab
- Group Discussion:
  - Discuss 2~4 mins

Textbook



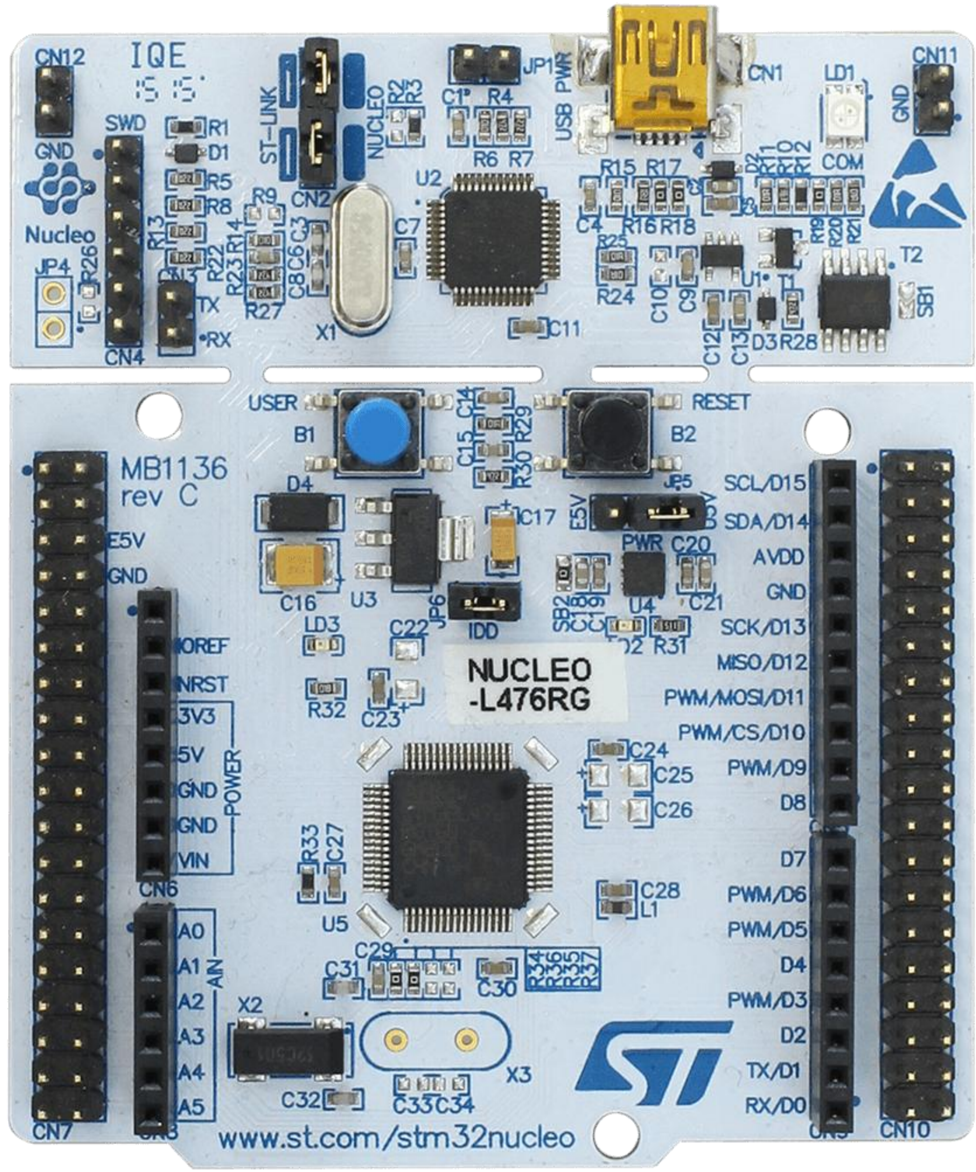
http://web.eece.maine.edu/~zhu/book/index.php

Labs

- There will be several (~6) labs (starting this week)
- Labs will deal with ARM based boards.
  - Lab projects can be downloaded from Blackboard
- You are welcome to use the lab anytime its open, however, please be aware that other people are using the lab for other courses too.
- You can minimize your experience in lab if you do some pre-lab work.
- From experience, it is advisable to get a composition notebook to write down everything handy about your lab and/or course work.

Lab Hardware and Software

- STM32 Nucleo-64 development board with STM32L476RG MCU (**Cortex-M4 with FPU and DSP**)
- You can buy it from [Mouser]
- **Software: Keil**


CN12
IQE
SWD
GND
Nucleo
JP4
CN4
TX
RX
ST-LINK
NUCLEO
CN2
X1
U2
C7
C11
JP1
USB PWR
CN1
LD1
COM
CN11
GND
T2
SB1
U1
D3
R28
USER
B1
RESET
B2
MB1136
rev C
E5V
GND
D4
C16
U3
C17
JP5
PWR
C20
U4
C21
JP6
IDD
LD3
R32
C22
C23
NUCLEO
-L476RG
C24
C25
C26
C28
L1
IOREF
NRST
3V3
5V
GND
GND
VIN
POWER
CN6
A0
A1
A2
A3
A4
A5
AIN
U5
X2
C32
X3
C33
C34
CN7
CN10
SCL/D15
SDA/D14
AVDD
GND
SCK/D13
MISO/D12
PWM/MOSI/D11
PWM/CS/D10
PWM/D9
D8
D7
PWM/D6
PWM/D5
D4
PWM/D3
D2
TX/D1
RX/D0
www.st.com/stm32nucleo

from st.com

STM32L4

Parallel Interface
FSMC 8-/16-bit
(TFT-LCD, SRAM, NOR, NAND)

Display
LCD driver 8 x 40

Timers
17 timers including:
2 x 16-bit advanced motor control timers
2 x ULP timers
7 x 16-bit-timers
2 x 32-bit timers

I/Os
Up to 114 I/Os
Touch-sensing controller

Cortex-M4
80 MHz
FPU
MPU
ETM

DMA

ART Accelerator™

Up to 1-Mbyte Flash with ECC Dual Bank

128-Kbyte RAM

Connectivity
USB OTG,
1x SD/SDIO/MMC, 3 x SPI,
3 x I²C, 1x CAN, 1 x Quad SPI,
5 x USART + 1 x ULP

Digital
AES (256-bit), TRNG, 2 x SAI, DFSDM (8 channels)

Analog
3 x 16-bit ADC, 2 x DAC,
2 x comparators, 2 x op amps
1 x temperature sensor

from st.com

6 Labs

1. Introduction
2. LED and pushbutton (in C)
3. Counting / 7-Segment Displays (in Assembly)
4. Keypad Scanning (in Assembly)
5. Stepper Motor Control (in Assembly)
6. Real-Time Clock Watch (in C)

Grade Breakdown

- Quizzes : 20%
  - In class
- Homework : 15%
- Midterm exam: 10%
- Final exam: 15%
- Labs: 40%

Homework

- Homework is accepted online.
- Homework will not be accepted if a student is late without a legitimate excuse.
- Homework will be returned in 2 weeks
- After homework is returned, you have 2 weeks to appeal grades. After that, it is firm.

Tests

- Tests are closed-book, closed notes.
- You can have a single-side cheat sheet.
- You must hand in this page of notes with your exam.
- I will review to help you preparing for exams one week ahead of the exam.

Statement on Classroom Recording

- To ensure the free and open discussion of ideas, students may not record classroom lectures, discussion and/or activities without the advance written permission of the instructor
- Any such recording properly approved in advance can be used solely for the student's own private use.

Diversity and Inclusion

- The University of Pittsburgh does not tolerate any form of discrimination, harassment, or retaliation based on disability, race, color, religion, national origin, ancestry, genetic information, marital status, familial status, sex, age, sexual orientation, veteran status or gender identity or other factors as stated in the University's Title IX policy.
- The University is committed to taking prompt action to end a hostile environment that interferes with the University's mission. For more information about policies, procedures, and practices, see: http://diversity.pitt.edu/affirmative-action/policies-procedures-and-practices.

Academic Integrity

- Please consult Academic Integrity website
  - http://www.provost.pitt.edu/info/ai1.html
  - Academic misconduct will result in a failing class for this class.
  - Please ask for help and do not cheat!
  - I am here to help you and I will go above and beyond to help you.
  - I am also here to help if you just need to talk.
- You can also read books on taking classes effectively!
  - This helped me tremendously in school!

Homework 0

- Write something about yourself that is impressive